

Circles and circle theorems - Set 2 - Solutions

Specification MM3.2, MM3.3, M5.8, M5.9.

Answer key

Question	Answer	Topic
1	D	Circles and circle theorems
2	A	Circles and circle theorems
3	B	Circles and circle theorems
4	F	Circles and circle theorems
5	C	Circles and circle theorems

Worked solutions

Question 1 CIR-02 • Circles and circle theorems • spec MM3.2

Find the complete set of values of k for which $x^2 + y^2 + 2kx - 4y + 3k + 4 = 0$ represents a circle.

Answer: D

$(x + k)^2 + (y - 2)^2 = k^2 + 4 - 3k - 4 = k^2 - 3k = k(k - 3)$. A circle needs a positive right-hand side: $k < 0$ or $k > 3$.

Question 2 CIR-05 • Circles and circle theorems • spec MM3.2

The line $y = x + c$ is a tangent to the circle $(x - 1)^2 + (y - 2)^2 = 8$. Find the possible values of c .

Answer: A

The distance from $(1, 2)$ to $x - y + c = 0$ is $\frac{|1 - 2 + c|}{\sqrt{2}} = \sqrt{8} = 2\sqrt{2}$, so $|c - 1| = 4$: $c = 5$ or $c = -3$.

Question 3 CIR-09 • Circles and circle theorems • spec MM3.2

Find the length of the tangent from the point $(7, 1)$ to the circle $x^2 + y^2 - 2x - 4y - 4 = 0$.

Answer: B

Centre $(1, 2)$, radius 3. Distance from $(7, 1)$ to centre: $\sqrt{36 + 1} = \sqrt{37}$. Tangent length $= \sqrt{37 - 9} = \sqrt{28}$.

Question 4 CIR-11 • Circles and circle theorems • spec MM3.2

Two circles have equations $x^2 + y^2 = 25$ and $(x - 8)^2 + y^2 = 9$. Which of the following statements is/are true?

- I The circles touch externally.
- II The circles intersect at exactly two points.
- III The point $(5, 0)$ lies on both circles.

Answer: F

Radii 5 and 3, centres 8 apart $= 5 + 3$, so the circles touch externally at one point (I true, II false). That point is $(5, 0)$: $25 = 25$ and $9 + 0 = 9$ (III true).

Question 5 CIR-16 • Circles and circle theorems • spec MM3.2

The point $P(x, y)$ moves so that its distance from $(3, 0)$ is twice its distance from $(0, 0)$. The locus of P is a circle. Find its radius.

Answer: C

$(x - 3)^2 + y^2 = 4(x^2 + y^2) \Rightarrow 3x^2 + 3y^2 + 6x - 9 = 0 \Rightarrow x^2 + 2x + y^2 = 3 \Rightarrow (x + 1)^2 + y^2 = 4$. Radius 2.